

let's not get into racism and racist stuff... fearful eyes? Well, The Muse is supposed to have sensitive eyes (though she does end up having fearful eyes, in the sense of being as dead-looking as they are). A "tremendous grinning" mouth? EveR-2 Muse has a sweet, lipstick mouth.

Much more importantly, the Steam Man was imagination, but EveR-2 Muse is a flesh-but-not-bled robot that exists.

There is, obviously, the "stronger and mightier" theme in the Steam Man: he possesses exceptional strength, so much so that he pulled a chariot in his role as Horse. Besides, he is 10 feet tall. Refer *Virtual Weapons In Real Life*, *Digit*, July 2007—the US now has an exoskeleton for soldiers that allows them to lift very heavy things, run very fast, and leap into the air like no-one leapt before.

Coming back, that steam-powered concoction was not called a robot; the first mention of the word was in the play *R.U.R.* (Rossum's Universal Robots) (English edition 1920) by a certain Karel Capek (don't blame him—he didn't name himself). Here, too, the robots are simply human-like machines. They also happen to be slaves, but badly-treated ones. Then someone gives them emotions, they revolt, kill most people, and take over. Bring on the clichés... but it wasn't a cliché in 1920!

No history lessons here, so we won't tell you why they were called "robots." Figure that one out. But we dug up this one about digging machines—that they feature in H G Wells' *War of the Worlds*—and here's one line: "So far as I could see, the (digging machine) was without a directing Martian at all." Meaning some directing was required, so the machine was self-directing... meaning it had some intelligence. That might be (in our opinion) the first mention of a "real" robot. This, in addition to the "taking over" theme, would involve some talk of AI. We'll come to that in a bit.

Androids are automatons that resemble human beings in the essential aspects, and so "android" is a generic term. But cyborgs are augmented in some way; they are super, well, super-something... either their senses are more acute, or they can live a thousand years, or something like that. But you already know, if you've been reading *Digit*. Refer a couple of our past issues.

Then take this one, from *Cloak of Anarchy*, 1972, by Larry Niven: "...when Ron was flush with



Biological evolution is too slow for the human species. Over the next few decades, it's going to be left in the dust"

Ray Kurzweil
Inventor and Futurist

money... he'd bought an implant-watch. He told time by one red mark and two red lines glowing beneath the skin of his wrist." Fascinating? No-one's done this, but we can. If a cyber-punk called Stelarc could implant an *entire ear* into his forearm (*Digit*, July 2007), why not a little watch with glowing hands? Stelarc's ear is clearly visible, meaning it's pretty close to the surface of the skin. A watch wouldn't be too hard. Ask a medical expert if you don't believe us. That guy went ahead and did it; you could if you wanted to. Not that you do.

Tanks, Logics, Metaverses

Now you know that in 1946, computers were huge and all that; they weren't networked; hell, they didn't even store information for other people to use. But in that year, someone called Murray Leinster was prophetic enough to write a story called *A Logic Named Joe*. Leinster is under-celebrated; in fact, he hardly is celebrated. Decide for yourself whether he should be—from this trimmed excerpt (our italics):

"You got a logic in your house. It's got keys instead of dials and you punch the keys for what you wanna get. It's hooked in to the tank... Say you punch 'Station SNAFU' on your logic. Relays in the tank take over an' whatever vision-program SNAFU is telecastin' comes on your logic's screen. Or you punch 'Sally Hancock's Phone' an' the screen blinks an' sputters an' you're hooked up with the logic in her house an' if somebody answers you got a vision-phone connection. But besides that, if you punch for the weather forecast, that comes on the screen too. The tank is a big buildin' full of all the facts in creation an' all the recorded telecasts that ever was made—an' it's hooked in with all the other tanks all over the country—an' everything you wanna know or see or hear, you punch for it an' you get it. Very convenient. Also it does math for you..."

First mention of the Internet? Could be. Computers in the house; connected to a server; podcasts; voice chat; a weather widget; Google servers that host much of the info we have access to; and servers connected to each other. And what else is the Internet?

1946! The mind boggles!

The Internet is quite something else when it comes to foresight: no-one knows what'll happen the day after tomorrow. *Snow Crash* by Neal Stephenson—1992—actually foresees *Second Life*!

"Hiro sees two young couples, probably using

The Present Of The Future: Featuring Ray Kurzweil, Futurist

Ray Kurzweil was fascinated by sci-fi in his childhood, and that's one reason we're talking about him. He invented many things, including OCR systems, a text-to-speech synthesizer, the CCD flatbed scanner, and more. Even so, he's now primarily known as a futurist—and here's some of what he says and what people think. (In 2005, *Gate* called Kurzweil "the best at predicting the future of artificial intelligence.")

☐ Kurzweil subscribes to the "Technological Singularity." This is the (supposed) creation of entities smarter than humans, which will accelerate technological progress beyond the capability of humans to participate in the progress. Think runaway bots doing stuff you can't understand. There are several criticisms we cannot go into,

but think of it as a hyper-extension of Moore's Law.

☐ Nanobots (which we've spoken about more than once in *Digit*) have the potential to help solve problems such as climatic change. They could also extend human lifespan significantly—even to the extent of immortality.

☐ (Another Herbert Simon?) Kurzweil says computers will become smarter than humans (crudely put) in 25 years from now.

☐ Elements of what you saw in *The Matrix* could very well come true in your lifetime.

Head to www.KurzweilAI.com for the complete picture. Such are the predictions of today: hedge your bets with confidence, because it'll happen (or it won't) in this life of yours.